

A Framework for Predictive Automotive Valuation

1. Title & Collection Method

A framework for predictive automotive valuation

Our dataset focuses on predictive automotive valuation. It is designed to model how a vehicle's secondary market value (price) is influenced by its structural characteristics and usage history. The dataset contains 60 samples and features a multi-variable architecture with 4 independent input features (X) mapping to 1 continuous numeric target (y)

2. Description of Features & Labels

This dataset contains vehicle's secondary market value.

- ✓ **Vehicle Model:** Fuses brand and body silhouette (e.g., Toyota SUV, BMW Coupe) to capture manufacturer equity and market tier.
- ✓ **Year:** Represents the production year, acting as a chronological metric for absolute depreciation.
- ✓ **Mileage:** Measures total distance driven, serving as a proxy for mechanical fatigue and wear.
- ✓ **Fuel Type:** Captures the vehicle's powertrain mechanics (Gasoline, Diesel, Hybrid, Electric), reflecting market demands driven by modern fuel economics.
- ✓ **Price (Target Label):** The estimated market value of the vehicle in USD.

3. Dataset Structure

The dataset consists of exactly 60 rows (samples) and 5 columns (variables).

Sample Table

Below is a 5-row slice of the raw, uncleaned dataset. It highlights the exact structural layout of the 5 columns alongside the deliberate, real-world data-quality anomalies (such as missing values, shorthand dates, string pollution, and mixed text casing) that must be resolved during preprocessing:

Vehicle_Model	Year	Mileage	Fuel_Type	Price
Ford Electric EV	2015	154320.0	electric	5000.0
Ford Coupe	2021	47,269	Elec	6400.0
BMW Coupe	2015	87219.0	Diesel	\$5000
BMW Electric EV	'18	59203.0	electric	13700.0
Hyundai SUV	2026	NaN	gasoline	5000.0

4. Quality Issue

A thorough exploratory data analysis (EDA) of the raw dataset reveals five distinct data-quality issues embedded across both the features and the target label. These issues represent common, real-world data collection anomalies that must be structurally resolved during the preprocessing phase before the dataset can be utilized by any machine learning algorithm:

- a. Missing data values
 - i. Numerical Features: the mileage column has completely blank entries (represented as NaN)
 - ii. Categorical Features: the fuel type column lacks data points for specific entries
- b. Mixed formatting and structural clashes (Data type)
 - i. In the Year column, chronological variables alternate inconsistently between standard 4-digit integers (2020), float syntax with unnecessary decimals (2021.0), and abbreviated text string shorthand's (such as 18 or 22)
- c. String pollution in numerical features

The mileage column contains text and formatting characters instead of clean integers or floats

 - i. It includes commas as thousands separators (47, 269), standard string suffixes (120k miles) , and descriptive text placeholders for unknown data (UNKNOWN)
- d. Categorical inconsistency

The text columns suffer from poor naming conventions, splitting single logical concepts into multiple redundant groups

 - i. Fuel type: the concept of a gasoline engine is divided into **Gasoline**, **Gas** and **gasoline**. Similarly, electric propulsion split into **Elec** and **electric**.
 - ii. Vehicle model: text boundaries contain erratic spacing (**BMW Coupe** with multiple hidden spaces) and inconsistent case error (lowercase **toyota suv**)
- e. Target variable pollution
 - i. It mixes raw numerical floats (**5000.0**) with embedded string currency characters (**\$19400** or **\$5000**)

- ii. Furthermore, some samples have completely missing target labels

5. The type of this problem (supervised or unsupervised)

This problem is classified as a **Supervised** Learning problem.

Justification

- a. Unsupervised Learning deals with unlabeled data; the algorithm is given a collection of features and is tasked with exploring the dataset to find hidden patterns, structures, or natural groupings (such as clustering similar customer profiles together) without any pre-defined correct answers.
- b. Supervised Learning relies on a structured dataset where every training sample consists of an input pair: a set of predictive features (X) and a corresponding, clearly defined output label (y). The algorithm's objective is to analyze these historical pairs, learn the underlying mathematical relationship between the inputs and outputs, and build a predictive function.

In our vehicle dataset, we have a clear, explicit target label (y): the Price column

6. Use case

- a. Machine learning applications
 - i. **Regressions:** this is the primary use case for the dataset because the target variable (price) is a continuous numerical value. Models like Linear Regression or Random Forest Regressors can be trained using features like Year and Mileage to predict the exact market value of a vehicle. This can be used by dealerships for automated trade-in evaluations.
 - ii. **Classification:** The dataset can be adapted into a classification problem by transforming the continuous Price column into discrete categories. For example, vehicles can be binned into "Budget" (under \$10,000) or "Premium" (above \$10,000). Alternatively, a model could predict a binary label like Is_Electric based on the vehicle attributes.
 - iii. **Clustering:** By completely removing the Price column, unsupervised learning algorithms like K-Means Clustering can look strictly at the independent features (Vehicle_Model, Mileage, Year, Fuel_Type) to find natural groupings in the data. This allows businesses to automatically discover hidden market segments, such as clustering high-mileage commercial trucks separate from commuter hybrids.
- b. Placement in the data science lifecycle

- i. **Data acquisition & understanding:** The process begins with generating or loading the raw, uncleaned vehicle CSV file to evaluate its initial structural shape (60 rows, 5 columns) and form baseline hypotheses.
- ii. **Data Preparation & Preprocessing:** This is the most critical phase for this specific dataset. Due to the intentional noise, this step involves stripping text characters (like \$ and ,), parsing inconsistent dates, standardizing categorical text naming, and using statistical methods (median/mode) to impute missing NaN values.
- iii. **Modeling & Training:** The cleaned dataset is transformed into a mathematical array and split into a Training Set (80%) to teach the regression model depreciation patterns, and a Testing Set (20%) to evaluate performance.
- iv. **Evaluation:** In this final stage, the model's price predictions on the hidden test set are calculated and verified using metrics such as R^2 Score (variance explained) and Mean Absolute Error (MAE) to check accuracy before deployment.